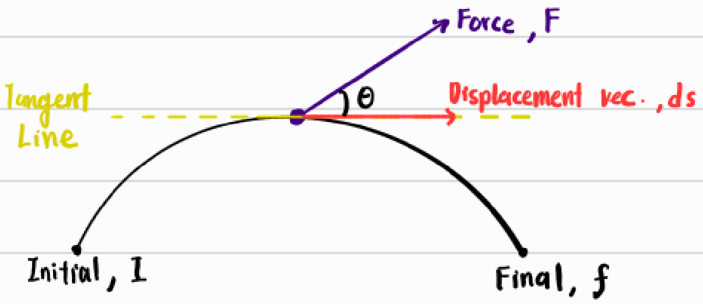
Work done by a force



Multiplication of 2 vector's magnitude

Work by F at that instantaneous point

$$W = \vec{F} \cdot \vec{ds}$$

$$= (\overbrace{F\cos\theta}^{F_x})(ds)$$

$$= F\,ds\cos\theta$$

Work by F from I to f

$$W_{\vec{F}} = \int_i^f \vec{F} \cdot \vec{ds}$$

Note :

If $\vec{F} \perp \vec{ds}$

$\Rightarrow \vec{F} \cdot \vec{ds} = F\,ds(\cos 90°) = 0$

$\therefore W_{\vec{F}} = 0$

Force classification

$\vec{F}$

- $W_{\vec{F}}$ independent of path taken → **Conservative**
  - Example : ① Weight
  - ② Spring force
- $W_{\vec{F}}$ dependent of path taken → **Non-conservative**
  - Example : ① Friction
  - ② Pull
  - ③ Push

Properties

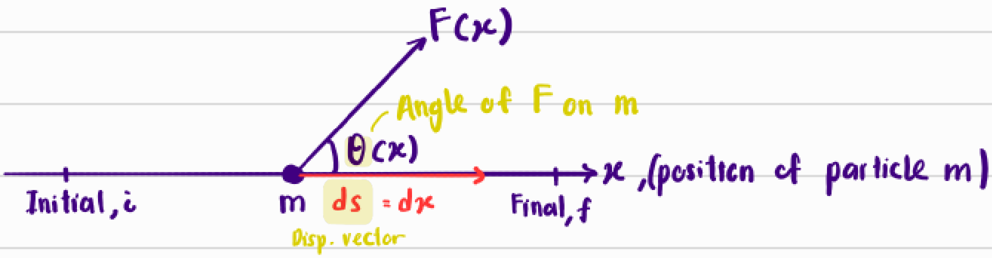
Force / Property	Consevative	Non - conservative
PE or KE conserved	✓	
Work independent of path taken	✓	
Work dependent of path taken		✓
Close - loop work = 0	✓	

Work done by a non-conservative force

Case ①: $\vec{F}$ is not constant

Direc. not constant
Magnitude not constant

Assumptions: ① Path is straight



Work done by $\vec{F}(x)$

$$W_{\vec{F}} = \int_i^f \vec{F}(x) \cdot ds$$

$$= \int_i^f \vec{F}(x) \cdot dx$$

$$= \int_i^f \{F(x) \cos[\theta(x)]\}\, dx$$

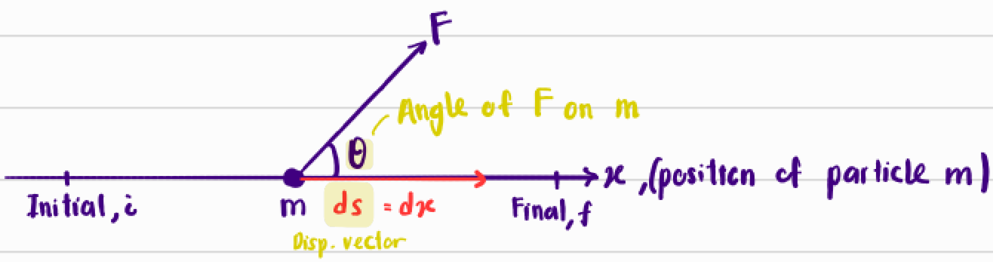
x-component of F(x) ⟹ $F_x(x)$

$$\boxed{W_{\vec{F}} = \int_i^f F_x(x)\, dx}$$

Case ②: $\vec{F}$ is constant

Direc. constant
Magnitude constant

Assumptions: ① Path is straight



Work done by $\vec{F}$

$$W_{\vec{F}} = \int_i^f \vec{F} \cdot ds$$

$$= \int_i^f \overbrace{F\cos\theta}^{F_x}\, ds$$

Since F is constant, θ is constant, the area below the graph of $F\cos\theta$ is a rectangle

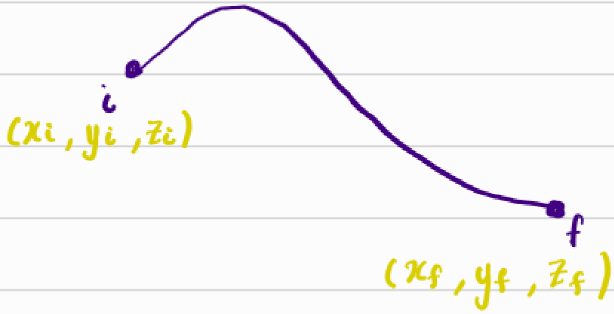
$$\boxed{\begin{aligned} \therefore W_{\vec{F}} &= (F\cos\theta)(d) \\ W_{\vec{F}} &= (F_x)(d) \end{aligned}}$$

Total dist.

If $\theta = 90° \Longrightarrow W_{\vec{F}} = 0$
$\theta = 0° \Longrightarrow W_{\vec{F}} = max = Fd$
$\theta = 180° \Longrightarrow W_{\vec{F}} = -Fd$

Work done by a conservative force

① Weight



$$W_{weight} = -mg[y_f - y_i]$$
$$= -mg\,\Delta y$$
$$= -\Delta U_g$$

$$\boxed{\begin{aligned} \text{Gravitational PE}, \ U_g &= mgy \\ \Rightarrow \Delta U_g &= mg\,\Delta y \end{aligned}}$$

② Spring force, F_{spring}

$$F_{spring} = -k(\underbrace{x - x_{equilibrium}}_{\text{Displacement}})$$
$$= -kx \quad [\text{Assume } x_{equil.} = 0]$$

$$W_{spring} = \int_{x_i}^{x_f} F_{spring}\,dx$$
$$= -\int_{x_i}^{x_f} kx\,dx$$
$$= -k\left[\frac{x^2}{2}\right]_{x_i}^{x_f}$$
$$= -k\left[\frac{x_f^2}{2} - \frac{x_i^2}{2}\right]$$
$$= -\left[\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right]$$
$$= -\frac{1}{2}k\,\Delta x^2$$

$$\boxed{W_{spring} = -\Delta U_e}$$

$$\boxed{\begin{aligned} \text{Elastic PE}, \ U_e &= \frac{1}{2}kx^2 \\ \Delta U_e &= \frac{1}{2}k\,\Delta x^2 \end{aligned}}$$

$\boxed{\text{Power}}$ Rate of change of work

$$\boxed{\text{Inst. power of } \vec{F} = \frac{d}{dt} W_{\vec{F}}(t)}$$

$$\boxed{\text{Avg power of } \vec{F} = \frac{W_{\vec{F}}}{t}}$$